

UNIVERSAL PLASTIC SHREDDER V2.0

Electrical Control & Power System

Bao Nguyen • Fearghus Tyler • Yaqoub Rabiah • Fox Kang

Portland State University • MCECS • Department of Electrical & Computer Engineering

Sponsor: MME / Ichor Systems (Bridget Lannigan) | Advisors: Prof. Andrew Greenberg, Dr. Jonathan Bird (EE), Robert Paxton (ME)



The Problem

Ichor Systems generates large volumes of thick rigid plastic (PVC, PP, LDPE) during semiconductor equipment manufacturing. Off-the-shelf shredders are either too small for the material (0.25–0.5 in. sheets, 4–12 in. wide) or priced far above the sponsor's per-line budget. The EE team designed and built the electrical control and power system for an industrial-grade plastic shredder; the mechanical structure is delivered by a separate ME team.

Requirements

Must:

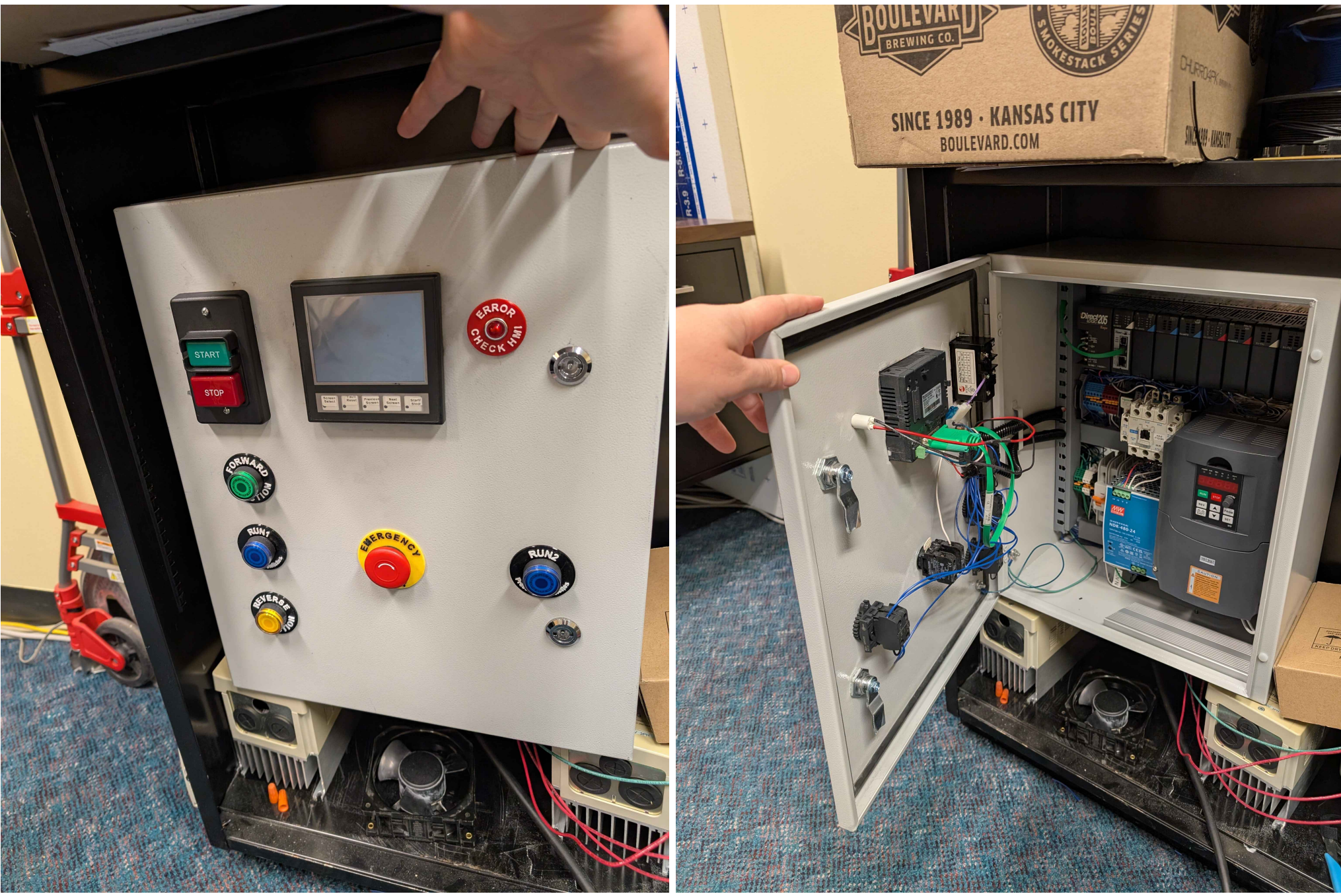
- Shred plastic to ~1 in. pieces.
- Drive a 3–5 HP motor at 70–90 RPM through a controllable VFD.
- Follow NFPA 70 (NEC); include E-stop, overload, and one safety interlock.
- Provide an HMI for operator control and status.
- Stay within the \$1,000 EE-team budget.

Should: status / fault lights, NEMA enclosure, labeled controls.

May: read-only wireless monitoring, data logging, capacitive-touch safety.

System Architecture

The control system is built end-to-end on AutomationDirect industrial hardware – no hobby-grade microcontrollers are in the safety or control path. A PowerTec 71008 disconnect feeds a Mean Well NDR-480-24 24 VDC supply, a Mitsubishi SD-N35 contactor (E-stop hardwired in series with its coil), and a Huanyang HY02D211B-T VFD. The VFD drives a 3 HP bench motor (GE 5KE182BC205B) over U/V/W via blue, white, and red 4 mm banana sockets.



Operator face (left), panel interior (right): HMI, deadman / fwd / rev / E-stop, DirectLOGIC 205, Huanyang VFD, 24 V supply, SD-N35 contactor, terminal blocks.

PLC Control

The PLC is the system controller. A DirectLOGIC 205 D2-09B-1 rack holds the H2-DM1E CPU and five I/O modules: D2-08ND3 (8 DC inputs), F2-08AD-1 (analog current input), F2-04RTD (RTD), D2-08TR (8 relay outputs), and F2-08DA-2 (0–10 V analog output to the VFD). The 31-rung Do-more program implements a deterministic state machine plus a 3-strike jam recovery sequence and a hard-lockout path.

HMI

AutomationDirect EA1-T4CL C-more Micro-Graphic 4 in. touch panel running HMI_413.mgp, serial to the H2-DM1E CPU. Four screens hold 27 PLC tags.



Motor Control, Motor Telemetry, Error / Alarm, Test.

Safety & Compliance

- E-stop:** Category 0 stop. Eaton E22B1 NC mushroom in series with the SD-N35 contactor coil. Drops the contactor independently of the PLC.
- Two-hand / deadman:** both pushbuttons must be held to enable the motor (PLC X2 AND X3).
- Overcurrent / jam:** VFD over-torque trip (PD124 = 150%, PD125 = 3 s) on PLC X5; ladder runs the 3-strike unjam routine.
- Enclosure:** Vorksmen SEB001 steel, NEMA Type 12.
- Compliance:** NFPA 70/79, NEMA 250 / ICS / MG 1, ISO 12100, 13849-1 (Cat B–1 / PL b–c), 13850, 13851, OSHA 1910.147.

Capacitive-Touch Safety (CTSI) Module

A contact-free operator-presence sensor designed and built in-house by the Shredder EE team. An ESP32 reads a capacitive electrode; on a touch event the ESP32 drives an LED through an optocoupler that closes the 24 V PLC input X7. Power is taken from the panel 24 VDC rail through an LM2596 buck converter to the ESP32's 3.3 V rail.

[CTSI module photo]
Drop Photo.CTSI.Module.jpg (or .png)
into 08.Poster/src/Pictures/ to insert it here.

ESP32 + LM2596 buck + optocoupler module wired to PLC X7.

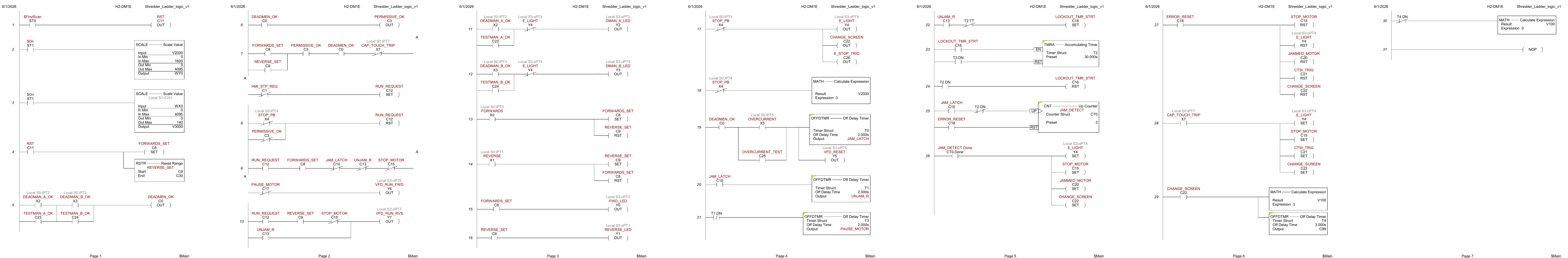
Ladder response (rung 28): touching the electrode latches Y4 (fault LED), C15 (STOP.MOTOR), C21 (CTSI.TRIG), and C22 (CHANGE_SCREEN) – so the motor stops, the panel fault indicator turns on, and the HMI jumps to the ERROR screen simultaneously and independently of the deadman / E-stop chain.

Status: bench-validated against the PLC; full machine-level verification on the enclosed prototype pending.

Results

- Panel built, wired, bench-validated. All 31 rungs program and scan correctly.
- All four HMI screens operate; tag database matches the PLC.
- Speed reference verified: HMI V2000 → SCALE → F2-08DA-2 CH1 → VFD VI.
- Jam-recovery verified: overcurrent on X5 → 2 s reverse (T1) → coast-to-stop buffer (T3).
- Three-revision point list signed off by Fearghus Tyler.

Complete Ladder Logic — 31 rungs across 7 pages



Pages 1–7: power-up init / deadman • RUN REQUEST gating / Y6 Y7 outputs • indicator LEDs / direction latches • E-stop / jam detection / unjam stroke • lockout timer / jam counter • fault reset / cap-touch / HMI screen change • screen reset / end of program.

Project By the Numbers

31 ladder rungs in production.

27 HMI tags across 4 screens.

14 component sheets in the point list.

3 strikes before the motor locks out.

2 s unjam reverse stroke.

150% over-torque threshold on the VFD.

Hardware Stack

PLC: DirectLOGIC 205 + H2-DM1E.

HMI: C-more EA1-T4CL.

VFD: Huanyang HY02D211B-T.

Motor: GE 5KE182BC205B (3 HP bench).

PSU: Mean Well NDR-480-24.

Contactor: Mitsubishi SD-N35.

Future Work

Production motor selection (3–5 HP) supersedes PD141--PD144.

Mechanical integration and shred-load run pending ME team.

Capacitive-touch (CTSI) module on X7: machine-level verification pending.

Optional Modbus link to the VFD for richer telemetry.

Acknowledgments

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